Agile Development Process and Devops

(ISWE406L)

School of Computer Science Engineering and Information Systems

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By

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Under the guidance

of

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SET-5

1.Problem Statement:

Ensure donation are processed and recorded accurately.

2. System Architecture & Tech Stack

2.1 Architectural Overview

The System is designed using a Continuous Integration and Continuous Deployment (CI/CD) pipeline to ensure automation, reliability, and scalability. The system follows a structured workflow where development, testing, and deployment processes are executed automatically without manual intervention.

At its core, the system uses Java-based logic to analyze input text and calculate similarity scores using a word-based comparison approach. This logic is integrated into a DevOps pipeline to ensure that only validated code is deployed.

The process begins when a developer pushes code to GitHub, which triggers the Jenkins pipeline. Jenkins retrieves the code, builds the application using Maven, and executes JUnit test cases to validate duplicate detection and similarity thresholds. If any test fails, the pipeline stops immediately, ensuring code quality.

If the tests pass, the application is packaged into a JAR file and containerized using Docker. The Docker image is then pushed to DockerHub and deployed to a Kubernetes cluster. Kubernetes manages multiple instances of the application, ensuring high availability and scalability, while services provide external access.

This architecture minimizes manual effort, reduces errors, and enables fast and reliable deployment.

Architecture Flow

Developer → GitHub → Jenkins → Maven Build → JUnit Testing → Docker Build → DockerHub → Kubernetes Deployment → Running Application

3. Project Modules

Your project contains:

1. Plagiarism Detection Module
2. Similarity Calculation Module
3. JUnit Testing Module
4. CI/CD Pipeline Module
5. Docker Containerization
6. Kubernetes Deployment

Suggested Project Structure

Hello-World

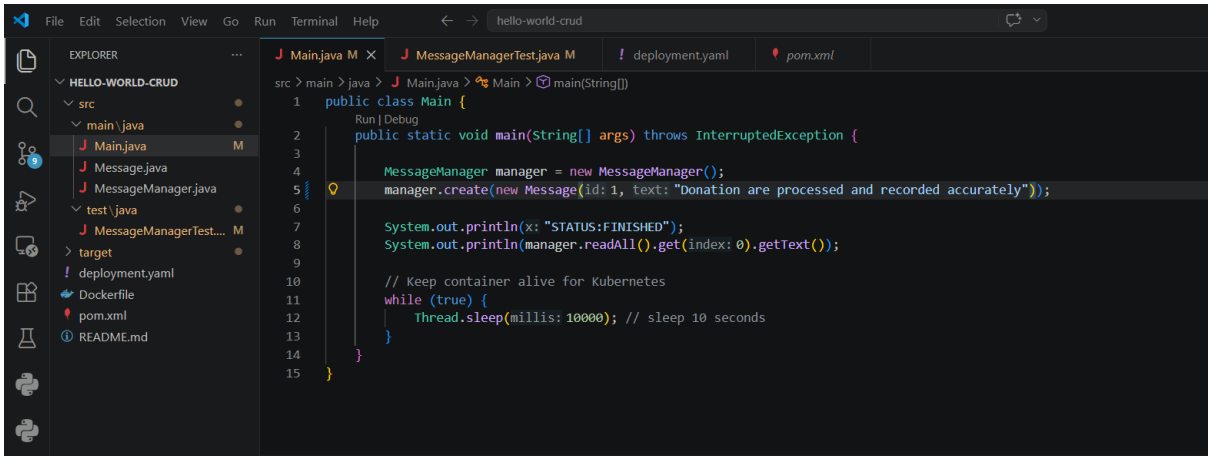
```
├──hello-world-crud/
├── src/
│   ├── main/java/
│   │   ├── Main.java
│   │   ├── Message.java
│   │   └── MessageManager.java
```

```
│   │
│   └── test/java/
│       └── MessageManagerTest.java
│
├── Dockerfile
├── deployment.yaml
└── pom.xml
└──README.md
```

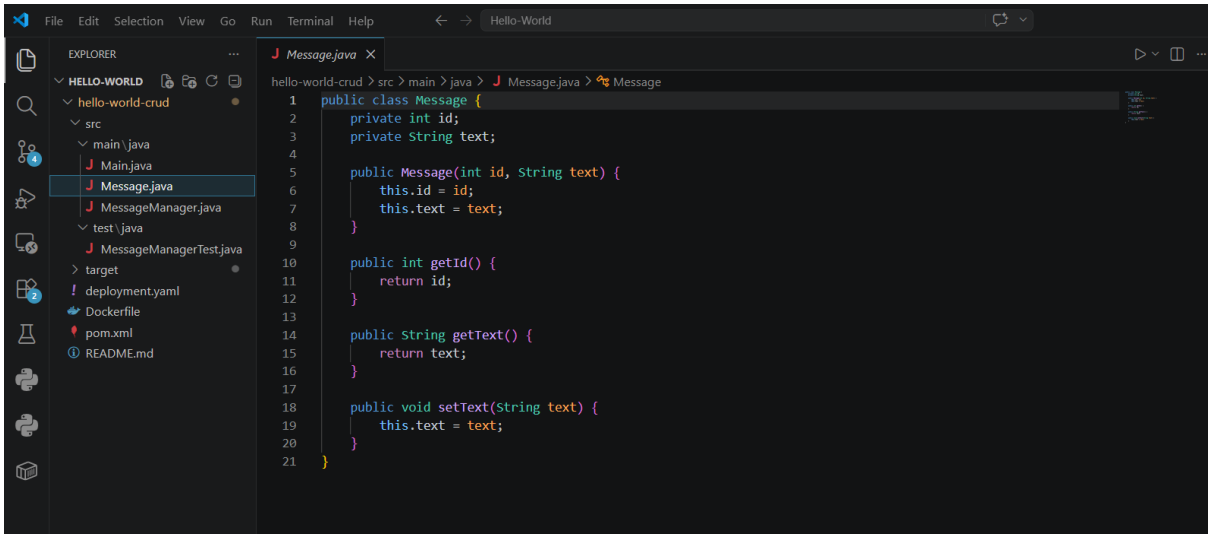
4. Implementation

4.1 Java Code :

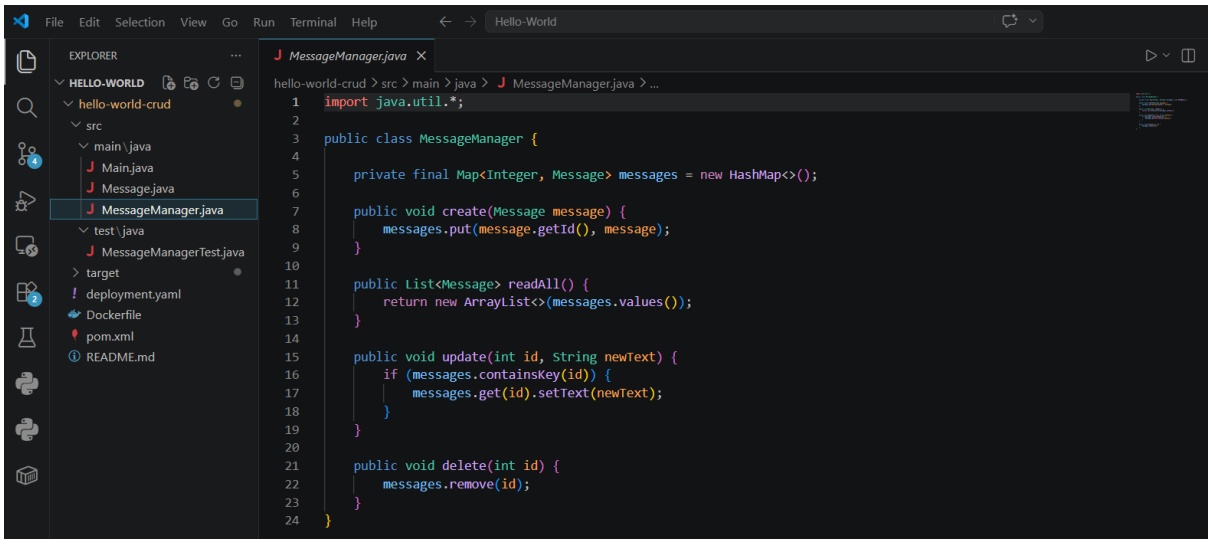
Main.java



Message.java

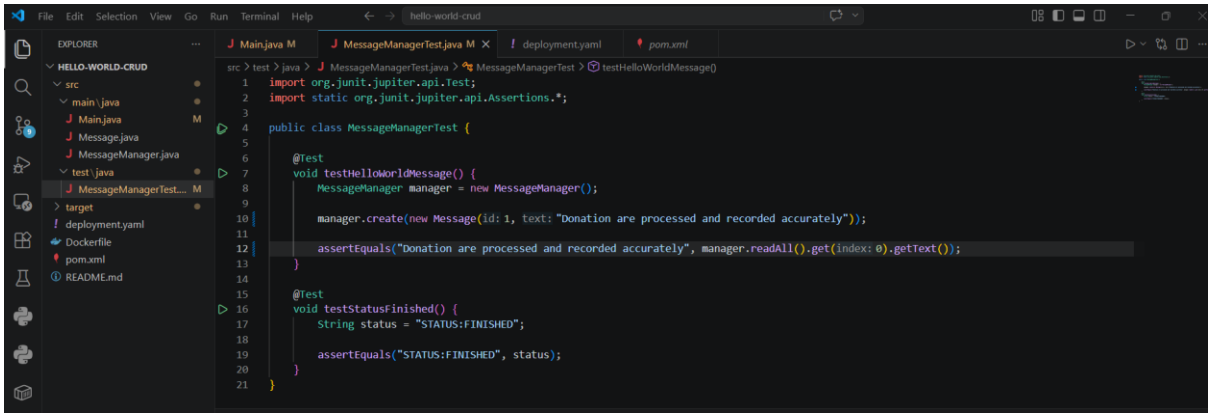


MessageManager.java

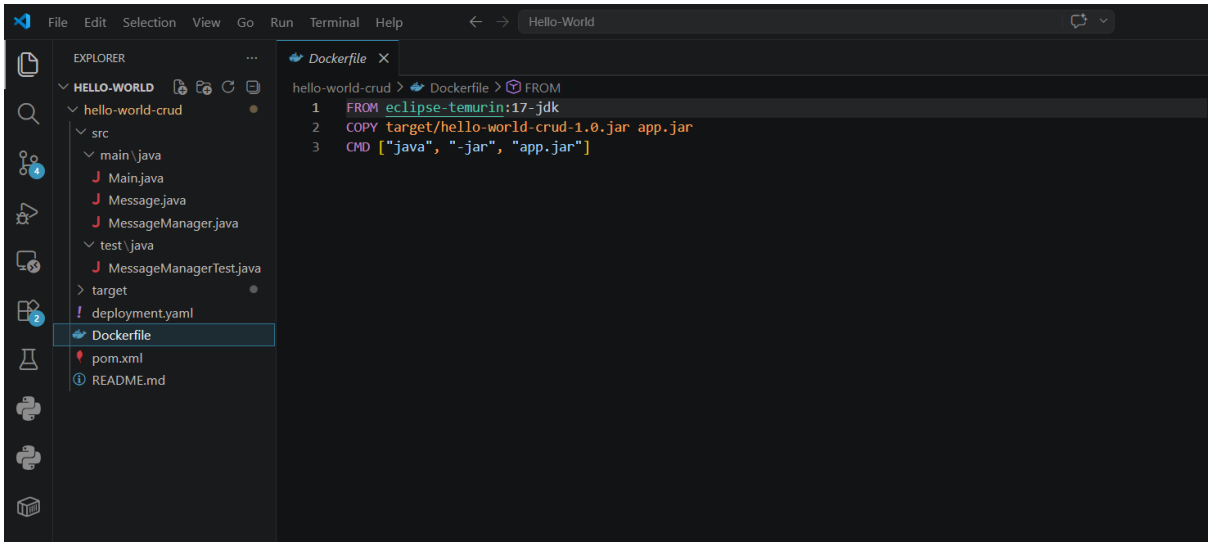


4.2 JUnit Test Cases

MessageManagerTest.java

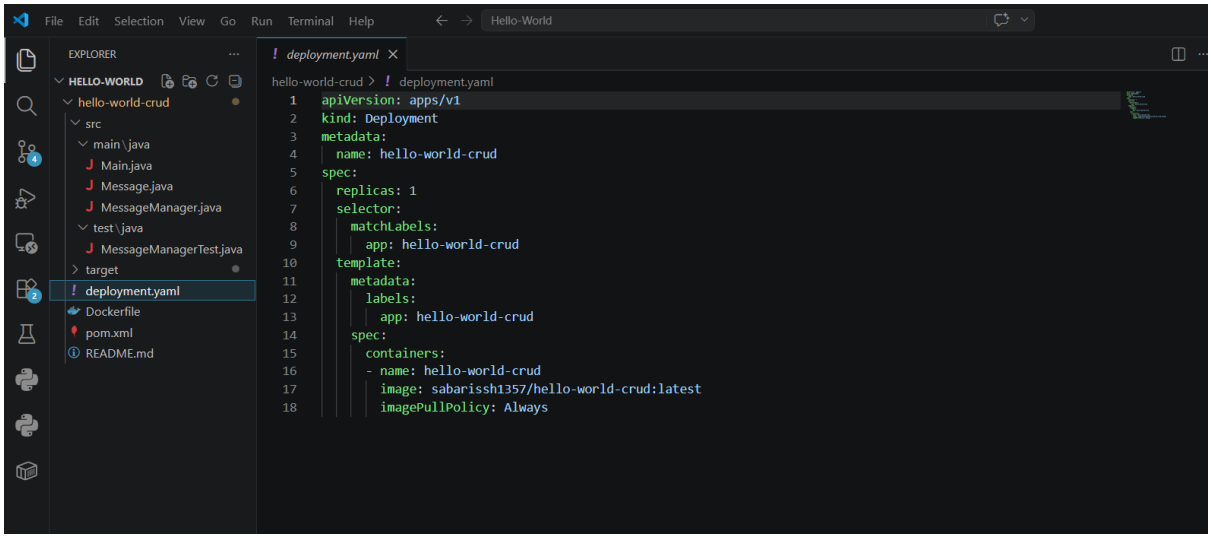


4.3 Dockerfile



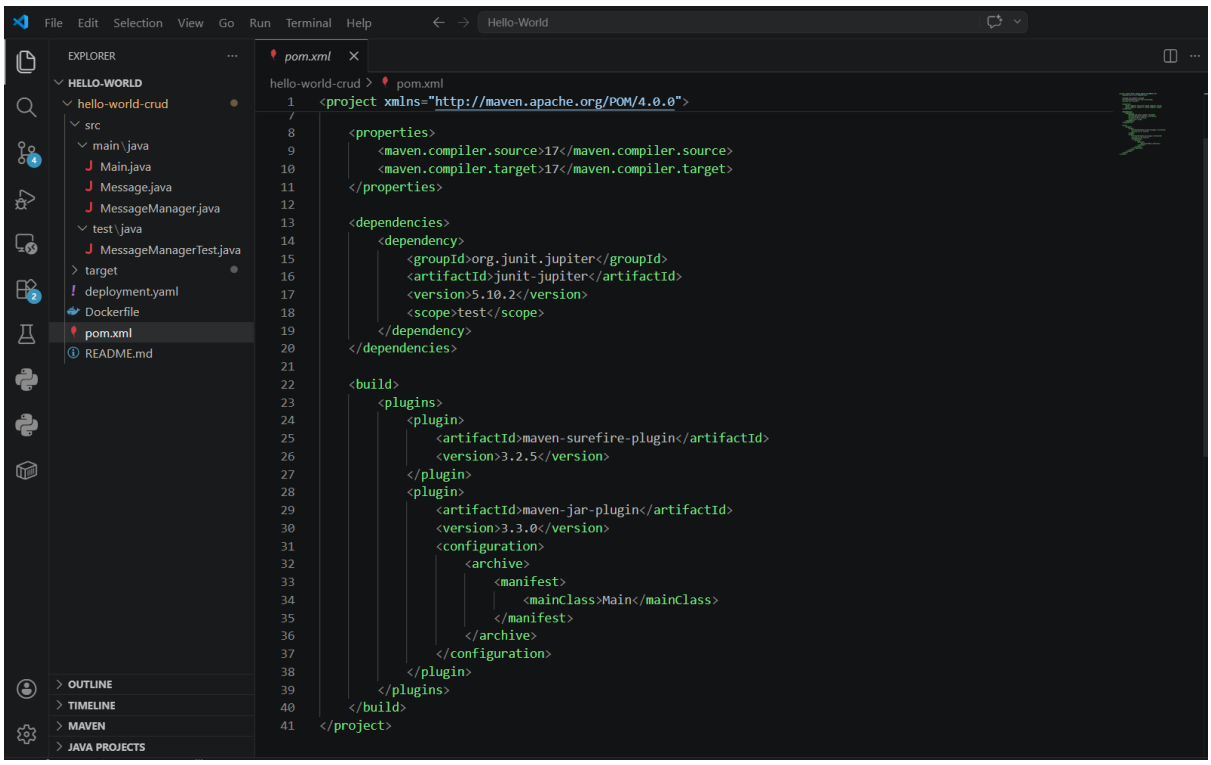
4.4 Kubernetes Deployment

deployment.yaml



4.5 Maven Configuration

pom.xml



4.6 Commands :

Use appropriate Git Commands to execute the following case studies in VS Code.

Build & Test (MAVEN):

Run Tests: mvn clean package

```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:hello-world-crud >-------------------
[INFO] Building hello-world-crud 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ hello-world-crud ---
[INFO] Deleting C:\Users\ADMIN\Documents\Hello-World\hello-world-crud\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ hello-world-crud ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\ADMIN\Documents\Hello-World\hello-world-crud\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ hello-world-crud ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ hello-world-crud ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
```

```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   java

[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running MessageManagerTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.270 s -- in MessageManagerTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.3.0:                                    world-crud ---
[INFO] Building jar:   Open file in editor (ctrl + click)   ello-World\hello-world-crud\target\hello-world-crud-1.0.jar
[INFO] Building jar: C:\Users\ADMIN\Documents\Hello-World\hello-world-crud\target\hello-world-crud-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  24.848 s
[INFO] Finished at: 2026-04-15T18:10:49+05:30
[INFO] ------------------------------------------------------------------------
```

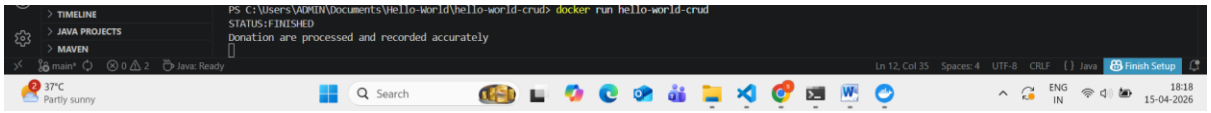
Build JAR: mvn clean package

```
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> java -jar target/hello-world-crud-1.0.jar
STATUS:FINISHED
Donation are processed and recorded accurately
```

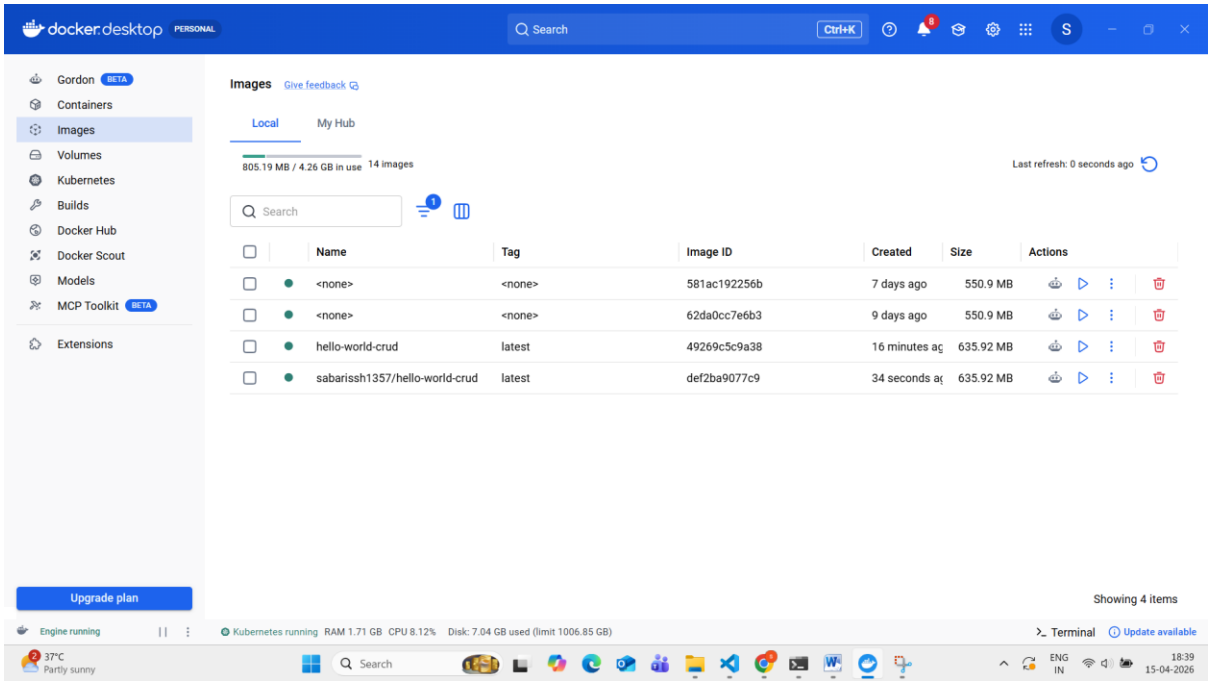
Docker Build

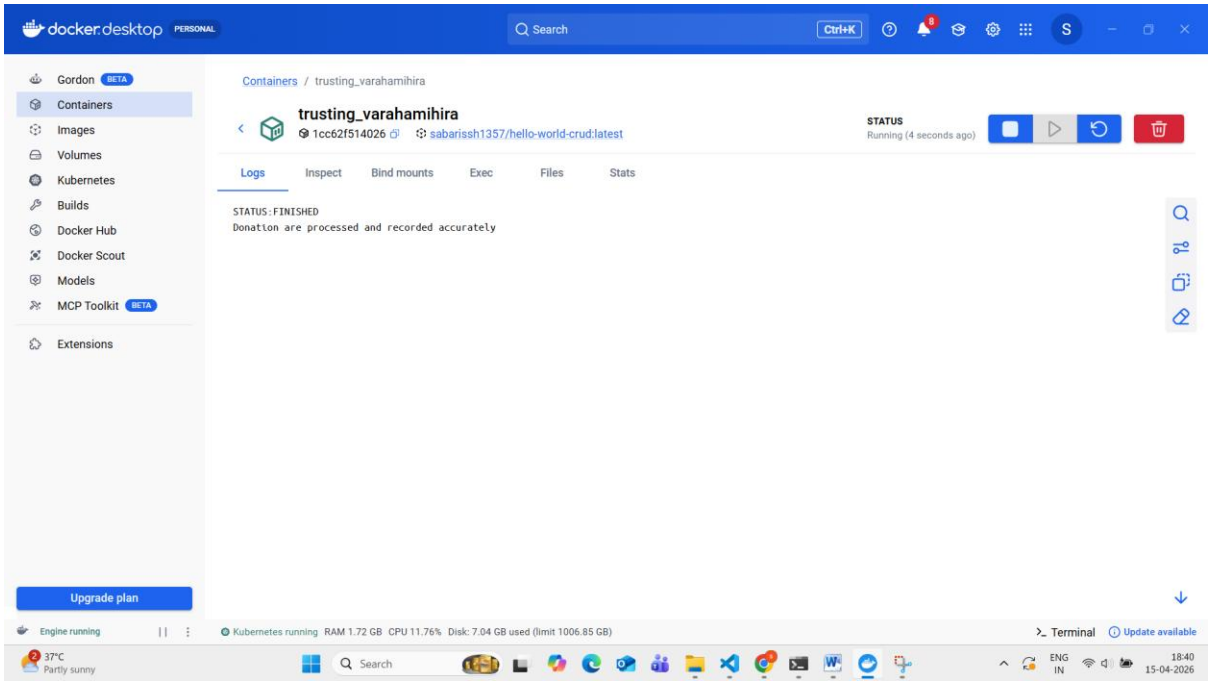
Cmd: docker build -t hello-world-crud

```
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> docker build -t hello-world-crud .
[+] Building 9.7s (8/8) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                              0.2s
 => => transferring dockerfile: 143B                                                                              0.1s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk                                         5.3s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                                            0.0s
 => [internal] load .dockerignore                                                                                 0.1s
 => => transferring context: 2B                                                                                   0.0s
 => [internal] load build context                                                                                 0.4s
 => => transferring context: 3.76kB                                                                               0.4s
 => CACHED [1/2] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:321764742e7c00b057cfdd36656bd45480d0a84a88cd77a5f707d6c99bcf1430  0.1s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:321764742e7c00b057cfdd36656bd45480d0a84a88cd77a5f707d6c99bcf1430  0.1s
 => [2/2] COPY target/hello-world-crud-1.0.jar app.jar                                                            0.2s
 => exporting to image                                                                                            2.5s
 => => exporting layers                                                                                           1.2s
 => => exporting manifest sha256:8f2ed2e82b2d108fa376e8c6578cc73daead7c7a440e868d916cab93ad0294f0                 0.1s
 => => exporting config sha256:51934b4270144ad4c57ef8e7eb3ff62669e452621804fb7736dd387d3c45d48f                   0.1s
 => => exporting attestation manifest sha256:f8755d5ad5a4d20c4b207aa1f94857b98c2c1185d8c07aa5dad4b1d649c1aab3      0.1s
 => => exporting manifest list sha256:49269c5c9a388562b236a106f88e057098d31b26790654f511771af6f699762b            0.1s
 => => naming to docker.io/library/hello-world-crud:latest                                                        0.0s
 => => unpacking to docker.io/library/hello-world-crud:latest                                                     0.4s
```



Docker Images(docker):





Push to Docker Hub:

Push: docker push Sabarissh1357/hello-world-crud:latest

```
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> docker push sabarissh1357/hello-world-crud:latest
The push refers to repository [docker.io/sabarissh1357/hello-world-crud]
45528c3f6daf: Already exists
2d2c31c38ad7: Layer already exists
a1e6e1a167f1: Layer already exists
a7428f4f5c1d: Layer already exists
a149c623348d: Layer already exists
689b91d88a0f: Layer already exists
f00bd8c488bb: Layer already exists
latest: digest: sha256:1323832d29a843164f43d5c463ee9c74c26061aca7d810127e4ff40dcb7da60b size: 856
```

Kubernetes Deployment:

Apply YAML

Cmd: kubectl apply -f deployment.yaml

```
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> kubectl apply -f deployment.yaml
deployment.apps/hello-world-crud created
```

Check pods

Cmd: kubectl get pods

```
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
hello-world-crud-68467bf968-5vcms   1/1     Running   0          13s
```

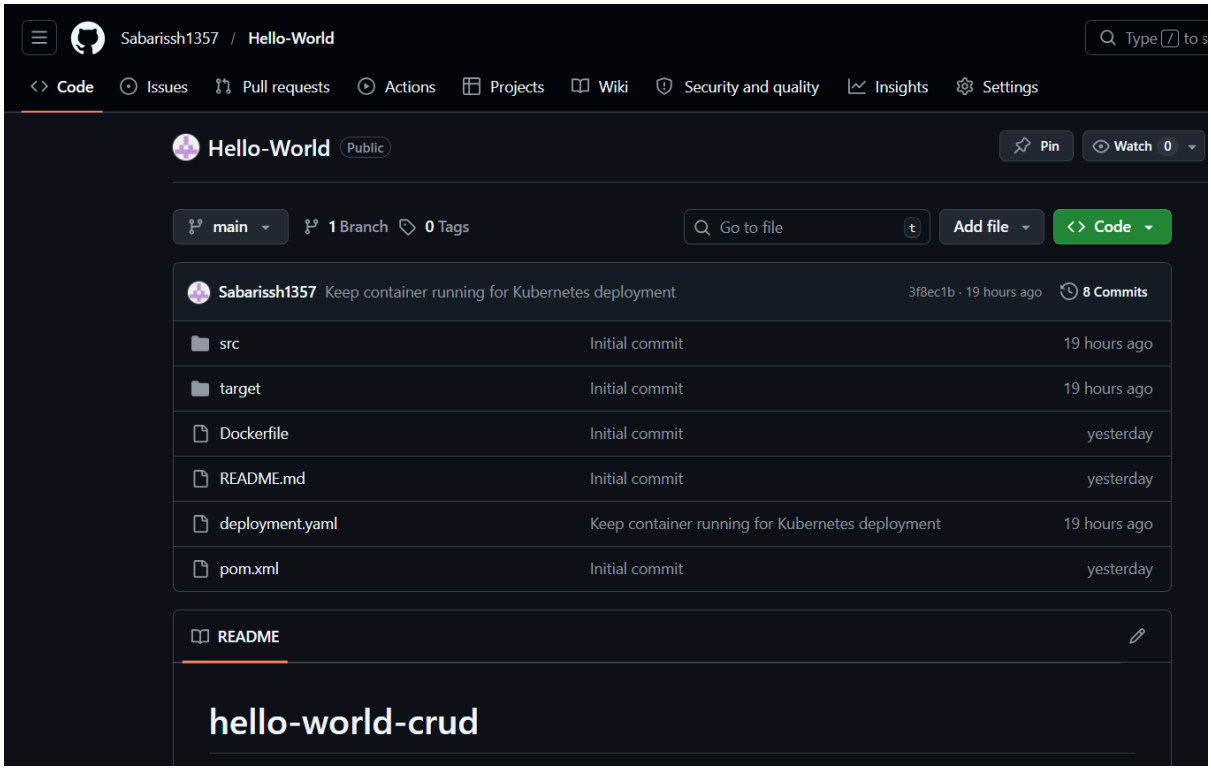
KUBERNETES(SERVICES):

Run Docker Container(Test):

Cmd: docker run hello-world-crud

```
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> kubectl apply -f deployment.yaml
deployment.apps/hello-world-crud unchanged
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> kubectl rollout restart deployment hello-world-crud
deployment.apps/hello-world-crud restarted
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> kubectl apply -f deployment.yaml
deployment.apps/hello-world-crud unchanged
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
hello-world-crud-9695cd995-ssp7s    1/1     Running   0          9s
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> kubectl logs hello-world-crud-9695cd995-ssp7s
STATUS:FINISHED
Donation are processed and recorded accurately
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud>
```

5. Git (For Jenkins Integration)




```
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> git init
Reinitialized existing Git repository in C:/Users/ADMIN/Documents/Hello-World/hello-world-crud/.git/
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst', LF will be replaced by CRLF the next time Git touche
s it
warning: in the working copy of 'target/surefire-reports/TEST-MessageManagerTest.xml', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> git commit -m "Initial commit"
[main 0433054] Initial commit
 5 files changed, 19 insertions(+), 19 deletions(-)
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> git remote add origin https://github.com/Sabarissh1357/Hello-World.git
error: remote origin already exists.
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> git remote remove origin
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> git remote add origin https://github.com/sabarissh1357/Hello-World.git
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> git branch -M main
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> git push -u origin main
```

```
PS C:\Users\ADMIN\Documents\Hello-World\hello-world-crud> git push -u origin main
Enumerating objects: 93, done.
Counting objects: 100% (93/93), done.
remote: Resolving deltas: 100% (20/20), done.
remote: This repository moved. Please use the new location:
remote:   https://github.com/Sabarissh1357/Hello-World.git
To https://github.com/sabarissh1357/Hello-World.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

6. Jenkins Pipeline Script

Script:

```
pipeline {
    agent any
```

```
environment {
    DOCKER_IMAGE = 'sabarissh1357/hello-world-crud:latest'
}

stages {
    stage('Clone Repository') {
        steps {
            git branch: 'main',
                url: 'https://github.com/Sabarissh1357/Hello-World.git'
        }
    }

    stage('Build Project') {
        steps {
            bat 'mvn clean package'
        }
    }

    stage('Run JUnit Tests') {
        steps {
            bat 'mvn test'
```

```
        }
    }

    stage('Build Docker Image') {
        steps {
            bat 'docker build -t %DOCKER_IMAGE% .'
        }
    }

    stage('Push Docker Image') {
        steps {
            withCredentials([usernamePassword(
                credentialsId: 'dockerhub-creds',
                usernameVariable: 'DOCKER_USER',
                passwordVariable: 'DOCKER_PASS'
            )]) {
                retry(3) {
                    bat 'docker login -u %DOCKER_USER% -p
%DOCKER_PASS%'
                    bat 'docker push %DOCKER_IMAGE%'
                }
            }
        }
```

```
        }

      stage('Verify Program Output') {
    steps {
        script {
            bat 'docker rm -f temp-verify-container || ver >nul'
            bat 'docker run -d --name temp-verify-container
sabarissh1357/hello-world-crud:latest'

            powershell 'Start-Sleep -Seconds 10'

            bat 'docker logs temp-verify-container | findstr "Donations are
processed and recorded accurately"'
        }
    }
}

        stage('Deploy to Kubernetes') {
            steps {
                bat 'kubectl apply -f deployment.yaml'
            }
        }
    }
```

```
    post {
        success {
            echo 'Deployment Successful'
        }
        failure {
            // Ensure cleanup happens even if the findstr check fails
            bat 'docker stop temp-verify-container || ver > nul'
            bat 'docker rm temp-verify-container || ver > nul'
            echo 'Pipeline Failed'
        }
    }
}
```

Console output:

SELECT:

Console

Download Copy View as plain text

```
Started by user Sabarissh S
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\ADMIN\.jenkins\workspace\hello-world-crud
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Repository)
[Pipeline] git
```

Jenkins / hello-world-crud / #3

```
C:\Users\ADMIN\.jenkins\workspace\hello-world-crud>docker run -d --name temp-verify-container sabarissh1357/hello-world-crud:latest
2ba3b10d6597eb68494f591b61588fc82ab2b8e0d95b5ef3f5c0b142ef2d09ed
[Pipeline] powershell
[Pipeline] bat

C:\Users\ADMIN\.jenkins\workspace\hello-world-crud>docker logs temp-verify-container  | findstr "Donation are processed and recorded accurately"
Donation are processed and recorded accurately
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] bat

C:\Users\ADMIN\.jenkins\workspace\hello-world-crud>kubectl apply -f deployment.yaml
deployment.apps/hello-world-crud created
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Deployment Successful
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.557

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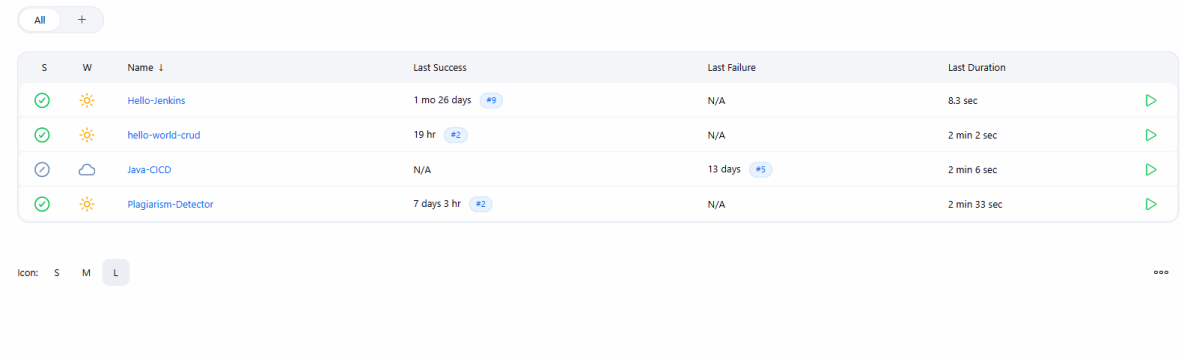
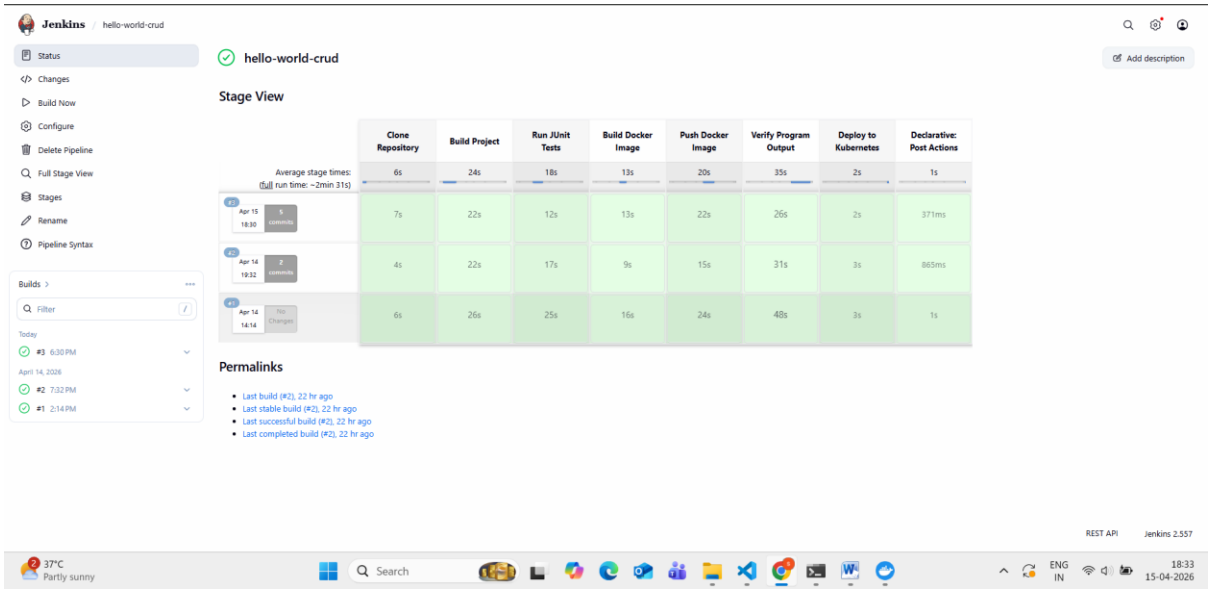
FINAL FLOW

1. Maven Build → mvn clean package
2. JUnit Testing → mvn test
3. Docker Image → docker build
4. Docker Push → docker push
5. Kubernetes Deploy → kubectl apply
6. Jenkins Automation → Pipeline

7. CICD Stage View:

TAKE PICTURE THIS AFTER COMPLETING



Stage View: